Chapter 2
Natural and Formal Systems

2.1 The Concept of a Natural System

In the present chapter, we shall be concerned with specifying what we mean by a *natural system*. Roughly speaking, a natural system comprises some aspect of the external world which we wish to study. A stone, a star, the solar system, an organism, a beehive, an ecosystem, are typical examples of natural systems; but so, too, are automobiles, factories, cities, and the like. Thus, natural systems are what the sciences are about, and what technologies seek to fabricate and control. We use the adjective "natural" to distinguish these systems from the *formal systems* which we create to represent and model them; formal systems are the vehicles for drawing inferences from premises, and belong to mathematics rather than science. There are of course close relations between the two classes of systems, natural and formal, which we hope to develop in due course.

In coming to grips with the idea of a natural system, we must necessarily touch on some basic philosophical questions, on both an ontological and an epistemological character. This is unavoidable in any case, and must be confronted squarely at the outset of a work like the present one, because one's tacit presuppositions in these areas determine the character of one's science. It is true that many scientists find an explicit consideration of such matters irksome, just as many working mathematicians dislike discussions of foundations of mathematics. Nevertheless, it is well to recall a remark made by David Hawkins: "Philosophy may be ignored but not escaped; and those who most ignore least escape". The discussion presented in this chapter is not intended to be exhaustive or comprehensive; it will, however, provide much of the basic conceptual underpinning for the detailed technical discussions of the subsequent chapters.[1]

It is perhaps best to begin our discussion of natural systems with our fundamental awareness of sensory impressions, which we shall collectively call *percepts*. These percepts are the basic stuff of science. It is at least plausible for us to believe that we do not entirely create these percepts, but rather discover them, through our experience of them. If we do not create them, then their source must lie, at least

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in part, outside of us, and thus we are led to the idea of an *external world*. We are further led to associate the sources of our percepts with definite *qualities* belonging to that external world; thence to identify such qualities with the percepts which they generate, and which we experience.

The first obvious step in our characterization of a natural system can then be stated: *a natural system is a member or element of the external world*. As such, then, it generates (or can generate, under suitable conditions) percepts in us, which we identify in turn with specific qualities, or properties, of the system itself.

The ability to generate percepts is not sufficient for us to say we have identified a natural system, although it is certainly a necessary part of such an identification. Intuitively, we would expect the concept of a system to involve some kind of inter-relation between the percepts it generates, and which then become identified with corresponding relationships between the external qualities which generated them.

The question of relations between percepts would be relatively simple if it were the case that such relations were themselves percepts. In that case, we could immediately extend our identification of percepts with qualities, so as to likewise identify perceptual relations with relations between qualities. However, we would not be justified in immediately taking such a step, because it is not clear that relations between percepts are themselves percepts or that they are discovered rather than created. This is a subtle and important point, which must be discussed in more detail.

Briefly, we believe that one of the primary functions of the mind is precisely to *organize* percepts. That is, the mind is not merely a passive receiver of perceptual images, but rather takes an active role in processing them and ultimately in responding to them through effector mechanisms. The *organization* of percepts means precisely the establishment of relations between them. But we then must admit that such relations reflect the properties of the active mind as much as they do the percepts which the mind organizes.

What does seem to be true, however, is the following: that the mind behaves *as if* a relation it establishes between percepts were itself a percept. Consequently, it behaves *as if* such a relation between percepts arises from a corresponding relation between qualities in the external world. Therefore it behaves as if such a relation between qualities in the external world were itself a quality, and as if the perception of this new quality consisted precisely of the relation it established between percepts. The basic point is simply that relations between percepts are, at least to some extent, creations of the mind which are then *imputed* to the external world. As such, they may be regarded as "working hypotheses", or, to use a more direct word, *models* of how the external world is organized.

Now we have already argued that the function of the mind, in biological terms, is to act as a transducer between percepts and specific actions, mediated by the organism's effector mechanisms. It is clear that these actions, and thus the models of the external world which give rise to them, are directly subject to natural selection. An organism which acts inappropriately, or in a maladaptive manner, will clearly not survive long. The very fact of our survival as organisms subject to natural selection thus leads us to suspect that the mechanisms of the mind for organizing percepts (i.e. for establishing relations between percepts) must have some degree of correspondence with objective relations existing between qualities in the external world which is, after all, the agency through which selection mechanisms themselves operate. This type of argument from selection, while obviously not a proof (and equally obviously, hardly an argument from first principles), makes it at least plausible that (a) relations between perceived qualities exist in the external world, and (b) that such relations, too, can be discovered by the mind. The process by which such relations are discovered, however, is not of the same character as the discovery of the qualities themselves; rather, it arises through the *imputation* of mentally created relations between particular percepts to the qualities which generate these percepts.

This discussion provides the second necessary ingredient in our characterization of a natural system. Specifically, we shall say that *a natural system is a set of qualities, to which definite relations can be imputed*. As such, then, a natural system from the outset embodies a mental construct (i.e. a relation established by the mind between percepts) which comprises a hypothesis or model pertaining to the organization of the external world.

In what follows, we shall refer to a perceptible quality of a natural system as an *observable*. We shall call a relation obtaining between two or more observables belonging to a natural system a *linkage* between them. We take the viewpoint that the study of natural systems is precisely the specification of the observables belonging to such a system, and a characterization of the manner in which they are linked. Indeed, for us *observables are the fundamental units of natural systems, just as percepts are the fundamental units of experience*.

To proceed further, we need first to discuss in more detail how qualities, or observables, are defined in general. We have already stated that a quality is a property of the external world which is, or which can be made to be, directly perceptible to us. Let us elaborate a bit on this theme.

The first thing to recognize is that our sensory apparatus, through which qualities in the external world are transduced to percepts in us, must themselves be part of the external world. That is, they themselves are natural systems of a particular kind, or at least involve such natural systems in an essential way. It is more precise to say that it is a specific interaction between a natural system and our sensory apparatus which in fact generates a percept. In fact, it must ultimately be a *change* or *modification* in the sensory apparatus, arising from such an interaction, which actually does generate the percept.

Furthermore, we know that we can also interact with the external world through our own effector mechanisms; thus these effectors themselves must project into the external world. Such interactions cause changes in the external world itself, which can then be directly perceived.

From these assertions, it is a small but significant act of extrapolation to posit that, in general, natural systems are capable of interactions, and that these interactions are accompanied by changes or modifications in the interacting systems. We have seen that, in case one of the interacting systems is a part of our own sensory apparatus, the vehicle responsible for its modification is *by definition* a quality, or observable, of the system with which that apparatus interacts. It is a corollary of these propositions that *if an interaction between any two natural systems* S_1, S_2 *causes some change in* S_2, *say, then the vehicle responsible for this change is an observable of* S_1, *and conversely*. That is, we wish to define observables *in general* as the vehicles through which interactions between natural systems occur, and which are responsible for the ultimately perceptible changes in the interacting systems arising from the interactions.

The above ideas are in fact crucial for science, because they indicate how we can discover new qualities of observables of natural systems, besides the ones which are immediately perceptible to us. More specifically, since our own effectors are themselves natural systems, they can be employed to bring systems into interactions which were not so before. In this way we can actively perform *experiments*, rather than being restricted to passive observation. Suppose for example that we are interested in some specific natural system S_1, some of whose qualities we can directly perceive. Suppose further that we bring S_1 into interaction with some other system S_2. Suppose that as a result of this interaction there occurs a directly perceptible change in S_2. We may then associate the change in S_2 with some specific quality or observable of S_1, which may (or may not) be *different* from those observables of S_1 which were directly perceptible to us. In this situation, we are in effect employing the system S_2 as a transducer, which renders the corresponding observable of S_1 directly perceptible to us. As we shall see shortly, this situation is the prototype for the idea of a *meter*, which not only defines the observable through which S_1 interacts with it, but actually enables us to *measure* this observable. And it is also clear that such considerations directly generalize the manner in which our own sensory apparatus generates percepts.

Let us return again to the situation above, in which we have placed a system S_1 into interaction with another system S_2. Let us suppose now, however, that the interaction results in a perceptible change in S_1. We may then say that, as a result of the specific interaction with S_2, a new *behavior* of S_1 has been generated. It is correct to say that this behavior is *forced* by the interaction with S_2, or that it is an *output* to the input consisting of the observable of S_2 through which the interaction occurs.

Both of these situations are fundamental to experimental science, and are the prototypes of the two ways in which we acquire *data* regarding natural systems. We shall have much to say about both of them as we proceed.

The above ideas have one further crucial ramification, which we now proceed to describe. This ramification is concerned with associating *quantity* with qualities or observables, and is implicit in the preceding discussion. Let us recall that we have *defined* an observable of a natural system S_1 as a capability for specific interactions of S_1 with other systems, such as S_2, and that such an interaction is recognized through a corresponding change in S_2. Let us imagine now that the system S_2 has the following properties: (a) any change in S_2 arising from interaction with another system S_1 can be associated with the *same* quality or observable of S_1; and (b) prior to the establishment of interaction between S_2 and some other system S_1, S_2 has been restored to an unchanged, fixed initial situation. The property (a) means that, though S_2 can in principle interact with many different systems S_1, the *agency* of S_1 responsible for a change in S_2 is a fixed observable; (b) means that the system S_2 is always the same before interaction is instituted.

Under these conditions, it may still happen that the particular perceptible change in S_2 arising from interaction with S_1 is different from that arising from interaction with another system $S_1{}'$. Since by hypothesis the *quality* responsible for the change in S_2 is in both cases the same, we can conclude that this quality may take on different *values*. In fact, we can *measure* the values of this quality by the differences between the changes arising in S_2. By this means, we introduce a new *quantitative* aspect into the characterization of natural systems; we may say that *if a quality of such a system corresponds to an observable, a quantity corresponds to a specific value of an observable*.

The essence of quantitative measurement, or the determination of the magnitude or value of an observable, is thus referred to the change caused through interaction with a specific system like S_2 in the above discussion. The role played by S_2 is then, as noted before, that of a *meter* for the observable in question; under the conditions we have stated, the possible changes which can be induced in S_2 can be identified with the set of possible values of the observable which induces them. Such an identification introduces a crucial notion: that the change in a meter for an observable can *represent*, or replace, the specific value of the observable which generated it. In other words, the change in a meter serves to *label* the values of the associated observable. This will turn out to be the crucial concept underlying the development of the subsequent chapters, when we turn to the concept of formal systems, and the representation of natural systems in terms of formal ones.

There now remains one more basic notion to be introduced before we can proceed with a more comprehensive discussion of the linkage relations which can obtain between observables of a system. This is the notion of *time*, which is of course central to any discussion of change, and hence is a fundamental ingredient in any discussion of interaction of natural systems of the kind sketched above.[2] For the present, we shall restrict ourselves to the notion of time as quality; we shall return to the matter in more depth in Sect. 2.3 below, which is devoted to dynamics.

The notion of time is a difficult and complex one. We may recall the plaintive words of Saint Augustine, who in his *Confessions* remarks: "What then is time? if no one asks me, I know: if I wish to explain it to one that asketh, I know not."

In accord with the spirit of the discussion we have presented so far, we must begin with our direct perception of time, or what is commonly called the "sense" of time. This is a percept, in the sense in which we have employed the term above. But it is apparently not a simple percept; we can discern at least two quite distinct and apparently independent aspects of our time perception. These distinct aspects must be interpreted differently when, as with all percepts, we attempt to associate them with specific qualities pertaining to the external world. These aspects are: (a) the perception of co-temporality or *simultaneity*; and (b) the perception of *precedence*, involving the distinction between past and future. It should be noted at the outset that both of these aspects of time perception involve *relations* between percepts. It clearly makes no sense to apply either of them to a single percept. For this reason, it does not seem appropriate to treat time directly as a *quality* or observable belonging directly to the external world. Indeed, we have interpreted all such qualities as potential capacities for interaction, as manifested especially in the moving of meters. It does not seem that either aspect of time perception is of this character.

On the other hand, we have the conviction that time can be "measured". Thus, if time is not to be regarded as a quality or observable, we must understand the sense in which it can be measured, or resolved into specific time values (instants). We shall begin with a consideration of this question.

The crucial concept here will be that of a *label*; the employment of qualities of one system to represent qualities of another. We have seen, for instance, that the values of an observable can be represented, or labelled, by the specific changes induced in other systems, or meters. Now we have argued that one essential aspect of our time perception expresses the *simultaneity* of other percepts. If this aspect truly reflects a relation between percepts, it follows from our earlier discussion that this relation involves a creative mental act, or model, about the external world which generated the percepts. It is at least plausible to infer that *the specific creative act of the mind involved here is the segregation of percepts into mutually exclusive classes, in such a way that percepts assigned to the same class are regarded as simultaneous*, *and percepts assigned to different classes are not simultaneous*. In other words, the mind imposes an equivalence relation, not on any specific set of percepts, but on the set of *all possible percepts*; it is this act of extrapolation which characterizes the creative aspect of the sense of simultaneity. Now in general, whenever we are given an equivalence relation on a set of elements, we may select a single representative from each equivalence class, and regard it as a specific *label* for that entire class. As we shall see later, the search for a clever choice of such a label or representative dominates all theories of classification; in mathematics, for instance, we search for *canonical forms* to represent classes of equivalent objects. If this is the case, we can conclude that a "time meter", or clock, is simply a natural system chosen in such a way that the particular percepts it generates serve as *labels* for the equivalence classes of simultaneous percepts to which they belong. Thus, a clock is not to be regarded as a meter at all, but rather as a generator of labels or representatives for these equivalence classes. In other words, a clock does not "measure" anything; it simply exemplifies the mentally constructed relation of simultaneity between percepts in a particularly convenient form. Its only relation to a true meter resides in its association of percepts with labels.

We may point out explicitly that there is no absolute or objective character to the classes of simultaneous percepts, or the corresponding classes of simultaneous qualities, which we have just described. Clearly, each of us creates these classes in his own way. We shall see later how we may use our basic time intuitions to construct an objective "common time", which all observers can use, in place of the subjective and idiosyncratic "time sense" which we have been describing.

We turn now to the other aspect of our time perception; that embodied in the distinction between past and future. This aspect imposes a different kind of relation on the set of possible percepts than that embodied in the sense of simultaneity; it is compatible with it, but apparently not directly inferable from it.

To describe this relation, which essentially expresses the idea of predecessor and successor, it is simplest to use a mathematical notation. Let t denote a particular instant; as we have seen, t is itself a percept belonging to the class $C(t)$ of all possible simultaneous percepts, and which serves to *label* the class. Let t' be another instant, $C(t')$ that class of simultaneous instants labelled by it. If we perceive that t' is later than t, then for each percept $p(t)$ in $C(t)$ there is a uniquely determined percept $p(t')$ in $C(t')$ such that $p(t)$ and $p(t')$ are related in the same way that t and t' are related. We alternatively say that $p(t)$ is a *predecessor* of $p(t')$, or $p(t')$ is a *successor* of $p(t)$. Thus, the relations of precedence in time serve to relate percepts belonging to *different simultaneity classes*. We must apparently posit separately that these precedence relations are such that the set of instants (i.e. the set of labels for the simultaneity classes) thereby acquires the mathematical structure of a totally ordered set; in fact, we shall see later that the natural mathematical candidates to represent such a set of instants are the integers or all real numbers.

If we take the now familiar step of imputing relations between percepts to corresponding relations in the external world, the above considerations suffice to characterize time in that world. However, simultaneity in the external world relates qualities or observables (or more precisely, the values of observables) and the set of instants serves to label classes of simultaneously manifested qualities; likewise, the relation of precedence serves to relate the qualities belonging to different simultaneity classes, again in such a way that the set of instants becomes totally ordered. We once again emphasize that these two relations reflect apparently different aspects of our sense of time, and it is important not to confuse them.

Let us now return to our discussion of natural systems, following this extensive detour into the character of observables and the nature of time. We recall that we had defined a natural system to be a collection of observables satisfying certain relations. We can now be somewhat more specific about the nature of the relations, or linkages, which may obtain between the observables of a natural system. To do this, we are going to utilize in an essential way the structures we have discussed in connection with time. Namely, we have described the sets $C(t)$ of simultaneous percepts, and the way these sets are related through the successor-predecessor relation. If we are given a natural system consisting of a set of related observables or qualities, we can then naturally consider:

(a) How these particular observables (or better, their specific values) are related as elements of a simultaneity class;
(b) How these observables are related through the successor-predecessor relation.

The first of these will specify how the values assumed by particular observables in the system at an instant of time are related to the values assumed by other observables. The second will specify how the values assumed by particular observables at a given instant are related to the values assumed by these or other observables at other instants.

In an intuitive sense, we would not say that a set of observables actually constitutes a system unless both these kinds of linkages can be specified, at least in principle. A direct expression of these linkages would be equivalent to specifying the *system laws*. For it is the function of system laws in science to tell us precisely how the values of the system observables at an instant of time are related, and how the values of these observables at a given time are related to their values at other times. As we shall see, the formal vehicles for expressing these laws are what we shall call *equations of state*. However, we shall argue that the precise formulation of such relations belongs not directly to the theory of natural systems, but depends in an essential way on the manner in which observables and their values are *represented* or *encoded* into corresponding formal systems. We will turn to such questions in the subsequent chapters of this section. For the present, it suffices to note that the existence of definite linkage relations of the kind we have described represent one embodiment of the notion of *causality*. Indeed, our initial insistence that definite linkage relations exist between the observables constituting a natural system restricts us from the outset to situations in which such a notion of causality is expressed. We shall later on see how this requirement can be loosened.

We wish to point out two other kinds of linkage relations which are implicit in the discussion of observables and measurement which was sketched above. However, whereas thus-far we have considered only linkages between the observables comprising a single system, the linkages we describe now relate observables belonging to *different* systems.

The first of these is implicit in the very concept of a meter. As we noted above, the essential function of a meter is to provide labels for the specific values of some observable, through the different changes arising in the meter when placed in interaction with a system containing the observable. This means precisely that the interaction between such a system and the meter establishes a relation, or linkage, between the observable being measured and some quality of the meter. Indeed, we shall see later that dynamical interactions provide the vehicles whereby linkages are established and lost in general; the case of the meter is a fundamental prototype for the modification of linkage relations. Thus in particular we can see how *the linkage relations themselves may be functions of time*. We note, for later reference, that the temporal variation of linkage relations arising from interactions will provide us with the vehicle for discussing phenomena of *emergence* and emergent novelty.

A final kind of linkage between observables belonging to different systems arises when, speaking roughly, two different natural systems comprising two different sets of observables, satisfy the same linkage relations. This shall provide us with the conceptual basis for the discussion of *system analogy*, and it is exemplified for example in the relation between a scale model and a prototype. We shall take up this kind of linkage further in our discussion of similarity.

Now let us pause for a moment and see where our discussion has led us so far. We have described the way percepts in us are associated with qualities in an external world. We have argued that such qualities, or observables, provide the vehicles for interaction between natural systems in that world. Such interactions can be exploited, through experiment, to make new qualities, initially imperceptible to us, visible. Moreover, such interactions provide us with a way of labelling the qualities of a system with changes in other systems; this is the essence of the concept of a meter. The importance of this concept lies not only in the fact that it allows us to *define* observables, but actually allows us to *measure* them, in the manner we have described. The exploitation of these interactive capabilities provide us with *data* about the external world; the acquisition of such data is the basic task of experimental science.

We also described certain kinds of relations between observables, which we called *linkages*. We saw how the establishment of such relations involved a constructive or creative aspect of the mind, expressed in the organization of percepts. The concept of time plays a crucial role in this organization of percepts; first in establishing a relation of simultaneity, and then in establishing a relation of succession or precedence. These relations are then imputed to the external world, just as qualities or observables are imputed to percepts.

Much beyond this we cannot go without introducing a fundamental new idea. The basic reason lies in the fact that experiment (i.e. the acquisition of data) can only tell us about observables and their values; it cannot in principle tell us anything about the manner in which these observables are linked or related. This is because relations do not directly generate percepts; they do not directly move meters. As we have seen, such relations are posited or *imputed* to the external world, and this imputation involves an essentially creative act of the mind; one which belongs to *theoretical* science. But how do we know whether such imputations are correct? Observation and experiment cannot directly answer such a question; all that can be observed, in principle, are observables and their specific values.

This is why a new element is needed before we can proceed further. Put succinctly, this new element must enable us to make *predictions* about observables on the basis of *posited* linkages between them. There is absolutely no way to do this within the confines of natural systems alone. What we require, then, is a new universe, in which we can *create* systems. But in this universe, the systems we create will have only the properties we decide to endow them with; they will comprise only those observables, and satisfy only those linkages, which we assign to them. But the crucial ingredient of this universe must be a mechanism for making inferences, or predictions. This capacity will play the same role in our new universe that causality plays in the external world. But whereas in the external world we are confronted with the exigencies of time, and are forced to wait upon its effects, in our new universe we are free of time; the relation of inference to premise takes the place of the temporal relation between predecessor and successor.

We have, of course, created such a universe. In its broadest terms, that universe is mathematics. The objects in that universe are *formal systems*. Hence, before we can go further in the universe of natural systems, we must explore the one of formal systems. We shall begin this exploration in the next chapter.

Looking ahead a bit further still, the next obvious task is to relate the worlds of natural systems with that of the formal ones; to exploit the capability of drawing inferences in the latter to make predictions about the former. This will be the subject of the final section in this chapter.

References and Notes

1. The epistemological considerations developed herein are idiosyncratic, but I think not arbitrary. I am sure that none of the individual elements of this epistemology is new, but I think that their juxtaposition is new and (judging by what it implies) significant. I see no purpose in contrasting my approach to others; this would at best be tangential to my primary purpose. For a fuller treatment of how the epistemological ideas developed herein arose out of definite scientific investigations, the reader is referred to

   Rosen, R., *Fundamentals of Measurement and the Theory of Natural Systems*. Elsevier, New York (1978).

2. For the purpose of the present discussion, it is sufficient to refer the reader to a general philosophical treatment of the problem of time, such as:
   Whitrow, G., *The Natural Philosophy of Time*. Nelson, London (1980).
   A more detailed discussion of how time in represented in different kinds of encodings will be found in the various Chap. 3 below, and the references thereto.

2.2 The Concept of a Formal System

The present chapter is devoted to developing the concept of a formal system. This development forces us to leave the external world of natural systems, with its observables which generate percepts and move meters. We must enter another world, populated entirely by mental constructs of particular kinds; this is the world of mathematics in the broadest sense.

This is not to say that the external world and the world of mathematics are entirely unrelated, or that our perception of natural systems plays no role in the construction of formal ones. As we shall see, there are many deep relations between the two worlds. Indeed, the main theme of the present book is the modeling relation, which rests precisely on linking the properties of natural systems with formal ones in a particular way. We wish merely to note here that mathematical objects, however they may initially be created or generated, possess an existence of their own, and that this existence is different in character from that of a natural one.

Furthermore, as we shall see, the world of mathematics really consists of two quite separate worlds, which are related to each other in very much the same way as the world of percepts is related to the external world. One of these mathematical worlds is populated by the familiar objects of mathematics: sets, groups, topological spaces, dynamical systems and the like; together with the mappings between them and the relations which they satisfy. The other world is populated with symbols and arrays of symbols. The fundamental relation between these two worlds is established by utilizing the symbols of the latter as *names* or *labels* for the entities in the former, and for regarding arrays of symbols in the latter as expressing *propositions* about the former. In this sense, the world of symbols becomes analogous to our previous world of percepts; while the world of mathematical objects becomes analogous to an external world which elicits percepts. The primary difference between the two situations (and it is an essential one) is that the mathematical worlds are entirely constructed by the creative faculty of the mind.

Perhaps the best way to illustrate the ideas just enunciated is to begin to construct the two mathematical worlds in parallel. We will do this by establishing a few of the rudiments of what is commonly called the *theory of sets*. This has been universally adopted as the essential foundation, in terms of which all the rest of mathematics can be built. As our purpose here is illustrative rather than expository, we shall suppose that the reader already has some familiarity with the concepts involved.[1]

Let us begin to populate our first world, which we shall denote by u. We will initially admit two kinds of entity into u, which we will call *elements* and *sets* respectively. We initially assume no relation of any kind between the entities called elements and those called sets.

Whenever we wish to speak of a particular *element*, we must characterize it by giving it a name; i.e. by labelling it with some symbol, say α, which serves to identify it, and distinguish it from others. This kind of labelling is very much like an act of perception; perceiving the particular element so labelled and its distinctness from others, which we labelled differently. Likewise, when we wish to speak of a particular *set* in u, we must provide it too with a name, or labelling symbol, say A.

Of course we must carefully distinguish the entities in and the symbols which we use to name these entities. The symbols themselves do not belong to u; only the entities themselves have this property. Their names thus must belong to another world, established in parallel with u; this world of names we shall call p.

As it stands, the world u is completely static and uninteresting. In order to proceed further, we must introduce some *relations* into it. We shall do this by admitting into our world a new kind of entity, neither a set nor an element. These new entities must also be labelled or named; let us agree to call such an entity (α, A), where α is the name of an element of u and A is the name of a set. Intuitively, we are going to interpret the presence of such a pair (α, A) in u by saying that α *is an element of* A.

At this stage, we can note that the *character* of u now depends entirely on which pairs are admitted into it. Thus already at this early stage we confront the necessity of *choice*; and indeed the choice we make here will determine everything which follows. Retrospectively, we can see that populating our universe u with different pairs will in effect create different (indeed, radically different) set theories, and hence different mathematics. We will return to this point several times as we proceed.

In accord with long tradition, we will agree to denote the presence of (α, A) in u by the introduction of a new symbol into p. This symbol serves to connect the names of the element and set constituting the pair; specifically, we will write

$$\alpha \,\varepsilon\, A \tag{2.1}$$

The symbol ε thus names a *relation* obtaining between sets and elements in u, and the entire expression (2.1) is interpreted as naming the manifestation of the relation between a particular pair in u.

The expression (2.1) is our first example of a *proposition*. Such a proposition does not belong to U, but rather is the name of an *assertion* about U. Since this is a crucial point, we will spend a moment discussing it.

Once we have introduced the symbol ε into P, there is nothing to prevent us from taking the name of an arbitrary element of U, say β, and the name of an arbitrary set, say C, and producing the expression or proposition

$$\beta \ \varepsilon \ C \tag{2.2}$$

This too is a *proposition*, and in effect it *asserts* something about U; namely, that (β, C) is a pair in U. But merely because the proposition (2.2) can be *expressed* in P, it does not follow that the relation it names actually *obtains* between β and C. Thus there are in general going to be *many more propositions in* P *than there are entities in* U.

If the particular proposition (2.2) happens to name a relation that actually obtains in U; (i.e. is such that (β, C) actually is in U) then we shall say that the proposition (2.2) is *true*. Otherwise, we shall say that (2.2) is *false*. Thus, the truth or falsity of such a proposition depends entirely on the nature of U; it cannot be determined from within the world P at all. In a sense, the characterization of (2.2) as a true proposition is an *empirical* question; we must look at U to determine if the particular pair (β, C) is in it. If not, then (2.2) is the name of a situation which does not exist in U; a label which labels nothing. But such a label cannot be excluded from P on any syntactical grounds; once the symbol ε is allowed, so are *all* propositions of the form (2.2). This is why, even at this most elementary stage, the world P of propositions has already grown larger than the world U which originally generated it. If we restrict ourselves only to the *true* propositions, then the two worlds remain in parallel. But as we emphasize, there is no way to do this within P. Indeed, the propositions (2.2) which now populate P express all the choices which *could have been* made in constructing U; they cannot tell us which ones actually *were* made.

It should also be stressed before proceeding that truth and falsity are attributes of *propositions*. It is therefore meaningless to speak of these attributes in connection with U; they pertain entirely to P alone. Basically, a false proposition is one which looks like a name, but which is in fact not a name (of anything in U).

Let us now proceed a bit further. The next step is, in effect, to change the rules of the game in U. What we shall do is to *identify* a set A in U with the totality of elements which belong to it. In other words, the entity originally named A completely disappears from view, and the name itself is reassigned to a particular totality of elements. This procedure can be represented in P with the aid of some new symbols introduced for the purpose:

$$A \equiv \{\alpha : \alpha \ \varepsilon \ A\}. \tag{2.3}$$

In words, A now names a particular family of elements of U; namely those for which the proposition α ε A (in the original sense) is true. The symbol "≡" thus expresses a *synonymy* in P, and will be exclusively used for this purpose. The notation in

(2.3) also serves to assign names to sets of elements α, characterized through the expression of a particular relation or property which they satisfy in u.

This identification makes it clear why the initial population of u which pairs (α, A) is so important. For at this stage, with the identification of A with the totality of its elements, the character of all subsequent developments in mathematics is determined. Indeed, the "best" way to do this is far from a trivial matter, and in fact is not really known. We do possess a certain amount of retrospective experience bearing on this question, but it is mainly of a negative character. For instance, if we populate u too liberally with such pairs, then some of our later set-theoretic constructions will turn out to have paradoxical properties; i.e. certain objects will need to be simultaneously included in and excluded from our universe. The well-known paradoxes of early set theory can all be regarded as arising in this manner.

Modulo the identification (2.3), we can now introduce all of the familiar relations and operations of elementary set theory. At each stage, we need to introduce corresponding *names* of these relations and operations into p. As we do so, we find exactly the same situation as we encountered with the introduction of ε; i.e. we are able thereby to form propositions in p which name no counterpart in u, and hence are to be considered *false*. In other words, as we build structure into u, we introduce a corresponding *syntax* into p, which generates a *calculus of propositions*.

To illustrate these last remarks, and incidentally to introduce some important terminology, let us consider the relation in u which expresses that a set A is a *subset* of a set B. By virtue of the identification (2.3) which specifies the sets of u in terms of their constituent elements, such a relation will devolve upon corresponding relations pertaining to the *elements* of A and B. The subset relation is said to obtain if u is such that, whenever a pair (α, A) is in u, the pair (α, B) is also; i.e. every element of A is also an element of B. Now to express this in p we need to relate the *propositions* $\alpha\ \varepsilon\ A$ and $\alpha\ \varepsilon\ B$. There is no way to do this with the machinery we have introduced into p so far; we need another *symbol* to express such a relation between propositions, just as we needed the new symbol ε to express a relation between names. This new symbol will be denoted by $\Rightarrow$, and will relate propositions in p, as in the expression

$$(\alpha\ \varepsilon\ A) \ \Rightarrow\ (\alpha\ \varepsilon\ B). \tag{2.4}$$

This expression is now a *new proposition*, which means: if $\alpha\ \varepsilon\ A$ is true, then $\alpha\ \varepsilon$ B is true. The subset relation can now be expressed in p, because if it obtains in u between a pair of sets A, B, the proposition (2.4) is true; conversely, if the proposition (2.4) is true for some A and B, we say that A is a subset of B. We still need a name for the subset relation itself; we thus write

$$A \subseteq B \equiv (\alpha\ \varepsilon\ A) \ \Rightarrow\ (\alpha\ \varepsilon\ B). \tag{2.5}$$

This name $A \subseteq B$ expresses now a relation between sets, instead of the oblique form (2.4) which relates propositions about elements. The relation named by (2.5) is called *inclusion*.

The implication relation ⇒ introduced above is going to be the central feature of our discussion of formal systems. As we shall see, it will be our primary vehicle for expressing in p all of the properties pertaining in u. Its basic characteristic is that *it never takes us out of the set of true propositions*; i.e. if p is any true proposition, and p ⇒ q holds for some other proposition q, then q is also a true proposition. Stated another way, this implication relation between propositions enables us to express *linkages* between properties in u; it will play the same role in the universe of formal systems that causality plays in the universe of natural ones.

Let us note in passing that the inclusion relation just introduced allows us to look at *synonymy* in a new light. For instance, if it should be the case that *both* of the inclusion relations

$$A \subseteq B, \quad B \subseteq A$$

hold in u, then both of the sets in question consist of exactly the same elements. By virtue of (2.3), then, these sets are not different in u, even though they bear two different *names*. Hence these names are synonyms for the *same* set in u. We shall soon see that much of mathematics consists of establishing that different names, or more generally, different propositions in p, are actually synonyms; in the sense that they name the same object or relation in u. The vehicle for establishing such synonymy will consist of chains of *implications* relating propositions in p; this is the essence of proof. We will take these matters up in more detail subsequently.

So far, we have considered how a set-theoretic relation (inclusion) is defined, and the manner in which it is accommodated in the world p of propositions, through the introduction of a corresponding relation between propositions. Let us now turn to the question of set-theoretic *operations*, using the specific example of the *union* of two sets.

In a sense, we can encompass a concept like the union of two sets under the rubric of set-theoretic relations, which we have already introduced. Intuitively, an operation like union defines a new set in a canonical fashion out of a pair of given sets. If C denotes this new set, we can equivalently define it in terms of the unique relation it bears to the sets from which it is built. That is, we can proceed by adding to our universe u a family of triples (A, B, C) satisfying the appropriate properties. For instance, we would informally require the set C in such a triple to be a subset of any set D of which A and B are both subsets.

However, it is more convenient (and more traditional) to proceed by specifying the new set C in terms of the specific elements of which it is composed, via (2.3). Intuitively, we wish an element α to belong to C in case α belongs to A, or α belongs to B, or belongs to both. To define C in this way requires us to articulate this idea as a *proposition* in p. We have no proposition at our disposal for this purpose in p so far, but we can *build* one if we allow ourselves a new operation on *propositions*. Of course, this is the *conjunction operation*, which from any two propositions p, q in p allows us to produce a new proposition denoted p v q. In terms of this operation on propositions, we can now specify the set we seek:

$$C \equiv \{\alpha : \quad (\alpha \, \varepsilon \, A) \vee (\alpha \, \varepsilon \, B)\}. \tag{2.6}$$

As before, we must give a specific name to the operation in u which constructs C for us out of A and B; we do this by *defining*

$$C \equiv A \cup B.$$

In words, the proposition $(\alpha \varepsilon A) \vee (\alpha \varepsilon B)$ is true precisely when at least one of its constituent propositions $(\alpha \varepsilon A)$ or $(\alpha \varepsilon B)$ is true. The expression $A \cup B$ is now to be regarded as the *name* of the set in u defined by (2.6).

Let us pause now to note a fundamental distinction between the world u of mathematical objects and the world p of names and propositions pertaining to these objects. In a certain important sense, a set like $A \cup B$ already exists in u from the outset; we need only find a way of giving it an appropriate name, which exhibits explicitly its relation to other sets in u. But the corresponding propositions in p need to be *created*, or *generated*; a proposition like $p \vee q$ does not exist in p before we introduce the specific operation "V", and directly apply it to particular propositions in p. This is a crucial point, and marks the difference between those mathematicians who believe that the mathematical universe pre-exists, possessing properties which must be *discovered*, and those who believe that the mathematical universe must be *constructed* or *created*, and that this construction is an entirely arbitrary process. Ultimately, this difference resides in whether one conceives of the natural habitat for mathematics as u or p. We need not regard these alternatives as being mutually exclusive, though perhaps most mathematicians have regarded them as such; this fundamental distinction, resting as it does primarily on aesthetic grounds, is responsible for much of the acrimony and bitterness surrounding the foundations of mathematics. But in any case we can discern once again the similarity of the relation between u and p to the relation between qualities and percepts which we discussed in the preceding chapter. We shall return to this matter when we consider the question of axiomatics below.

Let us return now to a consideration of the set-theoretic operation of forming the union. Once it has been defined or named, we can discover that it possesses certain properties. for instance that it is associative. Associativity consists in the fact that certain *names* of sets are in fact synonyms; specifically, that the two names

$$(A \cup B) \cup C \quad \text{and} \quad A \cup (B \cup C)$$

both label the same set in u. The *proof* of synonymy consists of establishing certain *implications* in p; namely, that

$$\alpha \varepsilon (A \cup B) \cup C \Rightarrow \alpha \varepsilon A \cup (B \cup C)$$

and

$$\alpha \varepsilon A \cup (B \cup C) \Rightarrow \alpha \varepsilon (A \cup B) \cup C$$

Thus, the sets named by $(A \cup B) \cup C$ and $A \cup (B \cup C)$ are each contained in the other; hence are identical. This identity is a *theorem*; an assertion which is true in u if the initial propositions

$$\alpha \ \varepsilon \ (A \cup B) \cup C$$

and

$$\alpha \ \varepsilon \ A \cup (B \cup C)$$

are true in u. We leave it to the reader to fill in the details. The reader should also convince himself that the associativity of the set-theoretic operation u *forces* the associativity of the operation V on propositions in p. The nature of this forcing is most illuminating in considering the relation between u and p.

We can proceed in an analogous manner to define the other basic set-theoretic operations of intersection (denoted by $\cap$) and complementation (denoted by —); these in their turn force us to introduce new operations on propositions in p (namely disjunction, denoted by $\wedge$, and negation, denoted variously by $-$ or $\sim$). We will omit the details, which the reader can readily supply. These simple ideas already admit a rich calculus (or better, an algebra, which is usually associated with the name of the English mathematician Boole), in which one can prove many theorems of the character we have already described. For instance, we can establish the familiar de Morgan Laws

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

in u, and their corresponding analogs in p. The reader should establish the appropriate chains of implication in detail, and interpret their meaning in the light of the preceding discussion.

Before proceeding further, let us add a word about the intersection operation. Explicitly, in analogy with (2.6), the intersection of two sets A, B is defined as the set $A \cap B$ of elements which belong both to A and to B; i.e. for which the propositions $(\alpha \ \varepsilon \ A)$ and $(\alpha \ \varepsilon \ B)$ are true. But clearly there need be no elements α in u for which $(\alpha \ \varepsilon \ A)$ and $(\alpha \ \varepsilon \ B)$ are both true. In this case we say that A and B are disjoint. By convention, and so that the intersection operation shall be universally defined for all pairs of sets A, B, we introduce a new concept; that of the *empty set*. The empty set intuitively possesses no elements, and is usually denoted by ϕ. Thus, if A and B are disjoint, we write

$$A \cap B = \phi.$$

The empty set is considered to be a perfectly respectable member of u (indeed, it can be regarded as generating all of u, in a certain ironic sense)[2], and is likewise considered to be a subset of every set; i.e. $\phi \subseteq A$ for every set A.

We now possess all the machinery for defining in u all of the familiar structures of mathematics. These can all be regarded as arising from particular kinds of set-theoretic operations. The most important of these set-theoretic operations is the *cartesian product* operation; given any pair A, B of sets we can define a new set $A \times B$ in the following way:

$$A \times B = \{(\alpha, \beta) \ : \ \alpha \ \varepsilon \ A, \ \alpha \ \varepsilon \ B\}.$$

An arbitrary subset $R \subseteq A \times B$ defines a *binary relation* between the elements of A and the elements of B. If R is such a relation, it is traditional to abbreviate the proposition $(\alpha, \beta) \varepsilon R$ by writing $\alpha R \beta$.

Binary relations formally lead us to some of the basic structures of mathematics, with which we shall be much concerned throughout the sequel. For instance, suppose that $R \subseteq A \times B$ is a binary relation. Suppose further that R possesses the following property

$$\alpha R \beta \quad \text{and} \quad \alpha R \gamma \Rightarrow \beta = \gamma.$$

A relation of this kind can thus be regarded as establishing a *mapping*, or *correspondence*, between the elements of A and the elements of B. Indeed, given a relation of this kind, we can express the proposition $\alpha R \beta$ by writing $\beta = f(\alpha)$, where f is the symbol expressing the mapping or correspondence. As we shall see, these mappings (also denoted by the symbol $f : A \to B$) introduce a dynamic element into U; they are the vehicles through which different structures in U may be compared.

One of the basic properties of binary relations, which we shall exploit heavily in subsequent chapters, resides in the fact that they can be *composed*. Suppose that $R_1 \subseteq A \times B$, and that $R_2 \subseteq B \times C$, are two binary relations. We are going to define a new relation $R \subseteq A \times C$. To define it, we need to specify what elements of $A \times C$ are in R. We shall say that a pair (α, γ) is in R if and only if there is a β in B such that

$$(\alpha, \beta) \varepsilon R_1 \quad \text{and} \quad (\beta, \gamma) \varepsilon R_2.$$

We shall define this relation R to be the *composite* of R_1 and R_2, and write $R = R_2R_1$.

In case R_1 and R_2 are *mappings*, it is easy to see that this definition produces a relation which is itself a mapping. This generates the familiar idea of *composition of mappings*. Specifically, if $f : A \to B$, and $g : B \to C$ are mappings, then there is a uniquely determined mapping $h : A \to C$ defined in accordance with the above construction, and we *define* $h = gf$.

These considerations lead directly to another set-theoretic construction, which is as important as the cartesian product. Namely, given two sets A, B in U, we can consider the set of *all* mappings from A to B. This is a new set, which we can denote by H(A, B). This set H(A, B) thus turns out to be intimately related to the cartesian product operation appropriately generalized, but we shall not develop that relation here.

If $A = B$, a binary relation $R \subseteq A \times A$ establishes a relation among the elements of A. By far the most important of these for our purposes are the *equivalence relations* on A. An equivalence relation $R \subseteq A \times A$ possesses the following properties:

(a) $\alpha R \alpha$ for every α in A (reflexivity)

(b) $(\alpha R \beta) \Rightarrow (\beta R \alpha)$ (symmetry)

(c) $(\alpha R \beta)$ and $(\beta R \gamma) \Rightarrow (\alpha R \gamma)$ (transitivity).

Such equivalence relations will be of vital importance to us throughout our subsequent development. Intuitively, an equivalence relation is a generalization of equality (which is itself obviously an equivalence relation) which arises when we discard some distinguishing property which enabled us to recognize that particular elements were unequal; when the property is discarded, such elements now appear indistinguishable.

This central aspect of equivalence relations is manifested in their fundamental property, which we can formulate as follows: if R is an equivalence relation on a set A, and if $\alpha \,\varepsilon\, A$ is any element of A, then we can form the subset of A defined as follows:

$$[\alpha] = \{\alpha' : \alpha \; R \; \alpha'\}.$$

If β is another element of A, we can likewise form the subset

$$[\beta] = \{\beta' : \beta \; R \; \beta'\}.$$

It is then a *theorem* that either

$$[\alpha] = [\beta].$$

or

$$[\alpha] \cap [\beta] = \phi$$

The subset $[\alpha]$ defined above is called the *equivalence class* of α (under the relation R); the theorem just enunciated asserts that two such classes are either disjoint or identical. Hence the relation R can be regarded as partitioning the set A into a family of disjoint subsets, whose union is clearly all of A. The *set* of equivalence classes of A under an equivalence relation R is itself a set in U, denoted A/R, and often called the *quotient set* of A modulo R. It is easy to see that the equivalence relation R on A becomes simple equality on A/R.

There is a simple but profound relation between mappings and equivalence relations. Let $f : A \to B$ be a mapping. For each $\alpha \,\varepsilon\, A$, let us define

$$[\alpha] = \{\alpha' : f(\alpha') = f(\alpha)\}.$$

These sets $[\alpha]$ are all subsets of A, and it is easy to see that they are either disjoint or identical. They can thus be regarded as the equivalence classes of a corresponding equivalence relation R_f on A. Intuitively, this relation R_f specifies the extent to which the elements of A can be distinguished, or resolved, by f. Thus, if we regard the mapping f as associating with an element $\alpha \,\varepsilon\, A$ a name or label $f(\alpha)$ in B, the equivalence relation R_f specifies those distinct elements of A which are assigned the same name by f. Moreover, it is a fact that *any* equivalence relation may be regarded as arising in this fashion. In this sense, equivalence is a kind of inverse of synonymy; equivalence involves two *things* assigned the same *name*; synonymy involves two *names* assigned to the same *thing*.

After all this extensive preamble, we are now in a position to discuss some familiar formal system as mathematical objects. In the world U, such a formal system can be regarded as a set together with some other structure; a set and a mapping; a set and a relation; a set and a family of subsets. As a simple example, we may consider a typical algebraic structure, such as a *group*.

A group may be regarded as a pair (G, τ), where G is a set in U, and $\tau : G \times G \to G$ is a definite mapping in H(G $\times$ G, G). This mapping τ is not arbitrary, but is supposed to satisfy the following conditions:

(a) $\tau(g_1, \tau(g_2, g_3)) = \tau(\tau(g_1, g_2), g_3)$ (associativity)
(b) $\tau(g, e) = \tau(e, g) = g$ for some distinguished element e ε G; the *unit element*
(c) for every g ε G there is a unique element g^{-1} ε G such that $\tau(g, g^{-1}) = \tau(g^{-1}, g) = e$.

It is more traditional to write $\tau(g_1, g_2)$ as $g_1 \cdot g_2$, and consider τ as defining a *binary operation* on G. The properties (a) to (c) above are usually called the "group axioms".

We can establish the framework for studying groups in p by mimicking the machinery which was established for sets, appropriately modified or constrained so as to respect the additional structure which distinguishes a group from a set. This machinery can be succinctly characterized under the headings (a) substructures, (b) quotient structures, (c) structure-preserving mappings. These are the *leitmotifs* which establish the framework in terms of which all kinds of mathematical structures are studied. For the particular case we are now considering, this *leitmotif* becomes: (a) sub-groups; (b) quotient groups; (c) group homomorphisms. Let us briefly discuss each of these in turn.

(a) *Subgroups*

As we have seen, a mathematical structure like a group is a set together with something else; in this case, a binary operation satisfying the group axioms. Thus a *subgroup* of a group G will intuitively first have to be a *subset* H $\subseteq$ G of the set of elements belonging to G. Further, this subset must bear a special relation to the binary operation, which is the other element of structure. It is clear what this relation should be: H should itself be a group under the same operation which makes G a group; and further should have the same unit element. A subset H with these properties is said to define a *subgroup* of G.

(b) *Quotient Groups*

To define quotient groups, we must first consider equivalence relations on G, considered as a set. But by virtue of the additional structure (the binary operation) we will only want to consider certain special equivalence relations, which in some sense are *compatible* with the additional structure. It is relatively clear how this compatibility is to be defined. Suppose that R is some equivalence relation on G. Let g_1, g_2 be elements of G. Suppose that g_1 R g_1', and that g_2 R g_2'. In G we can form the *products* $g_1 \cdot g_2$ and $g_1' \cdot g_2'$. It is natural to require that *these two products are also equivalent under* R; i.e. $(g_1 \cdot g_2)$ R $(g_1' \cdot g_2')$. If we do this, then the quotient *set* G/R can itself be turned into a group, by *defining* the product of two equivalence classes $[g_1]$, $[g_2]$ in G/R as

$$[g_1] \cdot [g_2] \;=\; [g_1 \cdot g_2].$$

That is: the product of equivalence classes in G/R is the equivalence class of the product in G. It is trivial to verify that this operation in G/R turns it into a new group; this new group is then called a *quotient group* of G.

(c) *Structure-Preserving Mappings*

Suppose that G and H are groups. Just as we did not wish to consider all equivalence relations on a group, but only those which were compatible with the additional structure, so we do not wish to consider all mappings $f: G \to H$, but only those which are also compatible with the structures on G and H. Just as before, this compatibility will consist in preserving the properties of the binary operations. Written succinctly, such compatibility means at least that

$$f(g_1 \cdot g_2) \;=\; f(g_1) \cdot f(g_2)$$

for any two elements g_1, g_2 in G. It should be noted that the binary operation appearing on the left side of this condition is the operation in G; the operation on the right side is the operation in H. We will also require explicitly that *f preserve units*; i.e. if e_G, e_H are the unit elements of G and H respectively, then $f(e_G) = e_H$. From this it immediately follows that such a mapping f also *preserves inverses*: $f(g^{-1}) = (f(g))^{-1}$.

A mapping $f: G \to H$ which satisfies these compatibility conditions is called a *group homomorphism*. Such a homomorphism is an instrument through which the group structures on G and H can be *compared*.

On the basis of these simple ideas, we can construct a world u_G patterned after our original world u, except that instead of consisting of sets it consists of *groups*; instead of allowing arbitrary mappings, we allow only group homomorphisms; instead of allowing arbitrary equivalence relations, we allow only compatible equivalence relations. Such a world is now commonly called a *category*; what we have done in specifying u_G is to construct the *category of groups*.

The above line of argument is generic in mathematics; it can be applied to any kind of mathematical structure, and leads to the construction of a corresponding *category* of such structures. Thus, we can consider categories of rings, fields, algebras, topological spaces, etc. Our original world u can itself be considered as a category; the *category of sets*. One of our basic tasks, which we shall consider in the next chapter, is to see how different categories may themselves be compared; i.e. how different classes of mathematical structures may be related to one another. In abstract terms, this kind of study comprises the *theory of categories*[3], and it will play a central conceptual role in all of our subsequent discussion.

In our discussion of formal systems so far, we have been concentrating on the world u. Let us now return to the world p of names and propositions, and see how such formal systems look in that world. As we shall see, the situation in that world is quite different. This is roughly because, as we pointed out before, we could essentially *discover* groups in u; but they must be *created* in p.

Let us then begin to create a group; the procedure we use will be typical of the constructive or generative procedures which characterize doing mathematics in p. We are going to initiate our construction by taking an array of symbols g, h, ... from p, now considered in the abstract; i.e. without regard for anything external to p which these symbols might name; we establish by fiat that these symbols shall be in our group. Now we must introduce a special rule which allows us to combine our symbols to obtain new ones analogous to the operations $\vee$, $\wedge$ etc., which allow us to combine propositions. We will call such a rule a *production rule*. The rule we use is a very simple one, called concatenation; if g, h are two of our symbols, then their concatenation gh will also be a symbol in our group-to-be. More generally, if we are given any finite number $g_1, \ldots, g_n$ of such symbols, then the concatenation $g_1 \ldots g_n$ is a symbol. In this way we build up finite *strings* of the symbols with which we started; such strings are conveniently called *words*. Most generally, if w_1, w_2 are words, then the concatenation w_1w_2 is also decreed to be a word.

Now concatenation is obviously associative: $(w_1w_2)w_3$ is the same string as $w_1(w_2w_3)$. Moreover, if we admit the *empty word* e into our system (where e is a string containing no symbols) then e plays the role of a unit element for concatenation; $we = ew = w$ for any word w. In this way, we have constructed in effect an algebraic system which has many grouplike properties; a binary operation (concatenation) which is associative and possesses a unit element. However, it is clear that there are no inverses in this kind of system, which is called the *free semigroup* generated by the symbols originally chosen (the significance of the word "free" will become clear in a moment).

To force the existence of inverses, we need to add them explicitly. Let us do this by taking another family of symbols $g', h', \ldots$, one for each of the generating symbols initially chosen, but such that the two families of symbols are disjoint. Intuitively, we are going to force g' to behave like g^{-1}, h' like h^{-1}, and so on. We do this by adding the new primed generating symbols to the original ones, and once again constructing all finite strings by concatenation. But now on this larger set of strings, we are going to *force* a concept of synonymy. In particular, we are going to decree that the string gg' and the string $g'g$ be synonymous with the empty word e; and likewise for all of the generating symbols. From this it is easy to see that, if $w = g_1 \ldots g_n$ is an arbitrary string, and if we define

$$w' = g_n{}' \ \ldots \ g_1{}',$$

then ww' and $w'w$ must also be synonymous with the empty word.

Intuitively, what this procedure does is to impose an equivalence relation R on our second set of strings; two such strings being called equivalent if one can be obtained from the other by the insertion or removal of substrings of the form ww' or $w'w$. It requires a bit of an argument to establish that the relation so introduced is indeed an equivalence relation, and that it is compatible with concatenation, but the details are not difficult and are left to the reader. Granting that this relation is a compatible equivalence, we can pass then to the quotient set modulo R. This quotient set, whose elements are *equivalence classes* of strings, has now all the properties of a group; it is called the *free group* generated by our original set of symbols.

The significance of the term "free" in these constructions is as follows: there is a sense in which the group we have constructed is the largest group which can be built from the generators we have employed. Stated another way: no *relation* is satisfied by the elements of this group, other than the one we have imposed to guarantee that they do indeed form a group.

If G is such a free group, then we can impose further (compatible) equivalence relations on G, and obtain new groups by forming the corresponding quotient groups G/R. It is not difficult to show that there is a sense in which *every* group can be regarded as a quotient group of a suitable free group, and in that sense, we have succeeded in constructing all groups in p. However, it is clear that the kind of constructive procedure we had to use to generate groups in p makes a group look much different from the simple definitions which could be employed to discover the same groups in u. The treatment adopted in the above constructive argument is often called *the definition of a group through generators and relations*. To construct any specific group, we must first construct a free group, and then impose on it precisely the relation R which ensures that the quotient modulo R is in fact the desired group. It may be imagined that this is not always easy to do[4].

We may note in passing that the free groups are in fact in u, but they are there characterized by the satisfaction of what appears to be a quite different property. For the reader's interest, we might sketch this property. Suppose that G is a group in u, and that A is simply a set. Let $f : A \to G$ be a mapping of A into G. Let H be any other group, and let $g : A \to H$ be any mapping. If in this case we can always find a group homomorphism $\phi : G \to H$ such that the composite mapping $\phi f : A \to H$ is the same as the chosen mapping $h : A \to H$, then G is called the free group generated by A.

In fact, the two definitions are equivalent, in the following sense. If we start with a set A in u, we can use the names of its elements as the symbols in p from which to begin our construction. When we do this, there is a direct *interpretation* of our construction in u at every stage, and in particular, the free group we generate names an object in u. This object can be shown to satisfy the condition just enunciated. In fact, this kind of argument is used to establish the *existence* of free groups. Indeed, by allowing our symbols to possess a definite interpretation in u from the outset, we may pass freely back and forth between p and u, exploiting both the constructive aspects of the former and the existential qualities of the latter. Mathematicians do this routinely. But it raises some deep questions, which we must now pause to examine.

These questions devolve once again on the relative roles played by u and p in the development of the mathematical universe. We have already touched on this above, and indeed, we have seen how differently the same mathematical object (a group) could look in the two realms. In general, we may say that those concerned with mathematics can be divided into two camps, or parties, which we may call the u-ists and the p-ists. The u-ists minimize the role played by p in mathematics; their basic position is roughly as follows: p if we wish to establish someproperty of a mathematical system, we need merely look at that object in u in an appropriate way, and *verify* the property. According to this view, mathematics is akin to an experimental science and, as noted before, u is treated in the same way an experimental scientist treats the external world.

The p-ists have, however, cogent arguments on their side. For one thing, all of the machinery of proof; i.e. for the establishment of chains of interference, is in p. Moreover, we can build into p an abstract image of any imaginable mathematical object, as a family of propositions generated from an appropriate set of symbols and production rules, whether u is available for inspection or not. Hence, they argue, u can be entirely dispensed with, for it can be invented entirely within p.

The u-ists can counter with a very powerful argument. Namely, as long as at least some of the symbols and propositions in p are interpretable as names of objects in u, the concept of truth or falsity of the propositions in p is not only meaningful, it is determined. Within p alone, there is no vehicle for assigning truth and falsity. Thus, since as we have seen p is inevitably much bigger than u, we will never effectively be able to find an abstract image of u in p. Moreover, the fundamental weapon of *implication*, which as we have seen never takes us out of a set of true propositions, is of no value without a *prior* notion of truth and falsity.

The p-ists can counter such an argument to some extent, as follows. We do not care, they may say, about any notion of truth or falsity which is imported into p from outside. From within p itself, we may by fiat declare a certain set of propositions to be *true*, together with all the propositions which may be built from these according to the syntactical rules in p. All we need care about is that this choice be made in such a way that the set of propositions arising in this fashion is *consistent*; in other words, that for no proposition p is it the case that both p and $-$p (the negation of p) belong to the set. And indeed, it may be argued that any such set of consistent propositions is an abstract image of *some* mathematical world u, and therefore *is* that world.

The p-ists have one more powerful argument, which is intended to rebut the idea that mathematical properties can be determined by inspection or verification in u. They point out that all mathematicians seek generality; i.e. to seek to establish the relation between mathematical properties in the widest possible context. Mathematicians seek to establish relations such as, “if G is *any* group, then some property p is true of G”. Or, stated another way, for *all* groups G, P(G) holds. In p, a proposition like this is represented through the introduction of the universal quantifier $\forall$, and would be expressed as

$$\forall G \ \ P(G)$$

Now the universal quantifier is closely related to the operation $\wedge$ of disjunction on propositions. For, if we had some way of enumerating groups, as say G_1, G_2, G_3, $\ldots$, G_n, $\ldots$, the above universal assertion could be written as

$$P(G_1) \ \wedge \ P(G_2) \ \wedge \ \ldots \ \wedge \ P\,(G_n) \ \wedge \ \ldots$$

To *verify* a proposition of this type requires an infinite number of verifications, which is impossible *in principle*. Thus, to the extent that a universal quantifier is meaningful in mathematics at all, it can only be meaningful within a limited constructive sense in a universe of totally abstract propositions.

The same argument may be given in connection with another kind of basic mathematical proposition, which asserts that for some particular property P, there *exists* an object A which satisfies the property. To enunciate such a proposition requires another quantifier, the *existential quantifier* ∃; in terms of it, the above assertion would be expressed as

$$\exists A \quad P(A).$$

This existential quantifier is closely related to the conjunction operation; again, if all objects A could be enumerated in some fashion as $A_1, A_2, \ldots, A_n, \ldots$, the existential assertion would amount to the proposition

$$P(A_1) \;\vee\; P(A_2) \;\vee\; \ldots \;\vee\; P(A_n) \;\vee\; \ldots$$

Here again, to establish existence by verification requires an infinite number of separate acts of verification, which is impossible in principle. Hence existence also can only be given a very circumscribed formal meaning, which involves the actual *construction* of the object whose existence is asserted.

Once again the u-ists can rebut in a very powerful way. Quite apart from the utter aridity of the manipulation of symbols devoid of interpretation, which is the nature of the universe contemplated by the p-ists, they can point out that the validity of the entire p-ists program would depend on *effectively* being able to determine the consistency of families of propositions arbitrarily decreed in advance to be true. It is here that the name of Gödel[5] first appears. To appreciate the full force of the argument we are now developing we must digress for a moment to consider the concept of *axiomatization*.

Axiomatization is an old idea, dating back to the codification of geometry by Euclid. The essence of the Euclidean development was to organize a vast amount of geometric knowledge by showing that each item could be established on the basis of a small number of specific geometric postulates, and a similarly small number of rules of inference or axioms. On this basis, it was believed for a long time that every true proposition in geometry could be established as a theorem; i.e. as a chain of implications ultimately going back to the postulates. Moreover, if some geometric assertion was not true, then its negation was a theorem.

Mathematicians came to believe that every branch of mathematics was of this character. Namely, they believed that the totality of true propositions about every branch of mathematics (say number theory) could all be established as theorems, on the basis of a small number of assumed postulates and the familiar rules of inference. It was this belief that in fact underlies the expectation that all of mathematics can be transferred from u to p; for the axiomatic method is entirely formal and abstract, and in principle utterly independent of interpretation. In this way, an axiomatic *theory* in p proceeds by identifying some small (invariably finite) number of propositions or symbols as axioms, and entirely by the application of a similarly small number of syntactical production rules (rules of inference) proceeds to generate the implications or theorems of the system. As noted above, the only limitation imposed here is that the resulting set of theorems be *consistent*; i.e. that for no proposition p are both p and −p theorems. The generation of theorems in such a system is entirely a *mechanical* process, in the sense that it can be done by a robot or machine; and indeed one approach to such systems proceeds entirely through such machines (Turing machines);[6] this is the basis for the close relation between axiomatics and automata theory. But that is another story.

What Gödel showed in this connection was that this kind of procedure is essentially hopeless. If we have some body of propositions that we desire to declare true, and if this body of propositions is rich enough to be at all interesting in mathematical terms, then there will be no mechanical process for establishing all these propositions on the basis of any finite subset of them chosen as axioms; i.e. there will be true propositions which are unprovable from the axioms, however the axioms may be chosen. There is only one exception; the fatal one in which the body of propositions initially declared true is inconsistent. Moreover, it is generally impossible, given a system of axioms and a set of production rules, to determine whether or not an arbitrary proposition is in fact a theorem; i.e. can be reached from the axioms by chains of implications utilizing the given rules.

Thus the u-ists can say to the p-ists: you claim that the impossibility of carrying out infinitely many acts of verification means that mathematics must become a game of manipulating meaningless symbols according to arbitrary rules. You say that truth or falsity of propositions is likewise something which may be arbitrarily assigned. But you cannot effectively answer any of the crucial questions; not even whether a proposition is a theorem, nor whether a particular set of propositions is consistent. Surely, that is no basis for doing mathematics.

And in fact, this is where the matter still stands at present. The dispute is over what mathematics *ought* to be; a question which would gladden Mr. Hutchins' heart, but one which cannot apparently be answered from within mathematics itself; it can only be answered from within the mathematician.

We are going to make the best of this situation by ignoring it to the extent possible. In what follows, we shall freely move back and forth between the world u of formal mathematical systems and the world p of propositions relating properties of these systems as if it were legitimate to do so; unless the reader is a specialist in foundation problems, this is in fact what the reader has always done. We will freely make use of *interpretations* of our propositions; in fact, the generation of such interpretations will be the heart of the modeling relation. The above discussion is important in its own right, partly because it teaches us the essential difference between things and the names of things. But also it provides some essential background for the discussion of modeling itself; many of the purely mathematical struggles to provide meaning through interpretation of sets of propositions are in themselves important instances of the modeling relation, and will be discussed in some detail in Chap. 3 below.

We now turn to the next essential item of business; the establishment of relationships between the worlds of natural systems and of formal ones.

References and Notes

1. Georg Cantor, the creator of set theory, envisioned it as providing an unshakable foundation for all of mathematics, something on which all mathematicians could agree as a common basis for their discipline. He would have been astounded by what has actually happened; hardly two books about "set theory" are alike, and some are so different that it is hard to believe they purport to be about the same subject. We have tried to make this liveliness visible in the treatment we provide. Of the multitude of books about set theory, the one whose spirit seems closest to ours is:

   van Dalen, D., Doets, H. C. and de Swart, H., *Sets: Naive, Axiomatic and Applied*. North-Holland (1978).
2. The empty set is perhaps the only mathematical object which survives Cartesian doubt; it must *necessarily* exist, even if nothing else does. Since it exists, we can form the class which contains the empty set as its single element; thus a one-element set also exists, and hence the number 1 can be defined. By iterating this procedure, we can construct a set with two elements, a set with three elements, etc., and mathematics can begin.
3. We shall discuss the Theory of Categories at greater length in Sect. 3.3 below; references are provided in the notes for that chapter.
4. In fact, it is generally impossible (in a definite, well-defined sense). At least, it is impossible in any sense that would be acceptable to a p-ist. The problem of determining effectively whether two words of even a finitely generated free group fall into the same equivalence class with respect to some relation is essentially the *word problem* for groups, and it can be shown to be unsolvable. See

   Boone, W. W., Cammonito, F. B. and Lyndon, R. C., *Word Problems*. North-Holland (1973).
5. In a celebrated paper, Gödel showed that any axiom system strong enough to do arithmetic must contain undecidable propositions, unless the system is inconsistent to begin with. Gödel's discovery devastated the so-called Hilbert Program, which sought to establish mathematical consistency entirely along p-ist lines. For a clear and simple exposition of Gödel's argument and its significance, we may recommend

   Nagel, E. and Newman, J. R., *Gödel's Proof*. Routledge & Kegan Paul, London (1962).
6. The term "machine" is used in mathematics for anything that executes an algorithm; it does not refer to a physical device. Such "Mathematical machines" provide an effective way to navigate in the universe p. The theory of such machines is known variously as recursive function theory and the theory of

automata. It also turns out to be intimately connected to the theory of the brain (cf. Sect. 3.5, Example 3B below) and cognate subjects. One way of showing that a class of problems (such as the word problems for finitely generated groups which was mentioned above) is unsolvable is to show that no machine can be programmed to solve the problem in a finite number of steps. General references bearing on these matters are:

Davis, M., *Computability and Unsolvability*. McGraw-Hill, New York (1958).

Minsky, M., Computation: *Finite and Infinite Machines*. Prentice-Hall, New York (1967).

2.3 Encodings Between Natural and Formal Systems

In the preceding two chapters, we have discussed the concepts of *natural systems*, which belong primarily to science, and *formal systems*, which belong to mathematics. We now turn to the fundamental question of establishing relations between the two classes of systems. The establishment of such relations is fundamental to the concept of a model, and indeed, to all of theoretical science. In the present chapter we shall discuss such relations in a general way, and consider a wealth of specific illustrative examples in Chap. 3 below.

In a sense, the difficulty and challenge in establishing such relations arises from the fact that the entities to be related are fundamentally different in kind. A natural system is essentially a bundle of linked qualities, or observables, coded or named by the specific percepts which they generate, and by the relations which the mind creates to organize them. As such, a natural system is *always incompletely known*; we continually learn about such a system, for instance by watching its effect on other systems with which it interacts, and attempting to include the observables rendered perceptible thereby into the scheme of linkages established previously. A formal system, on the other hand, is entirely a creation of the mind, possessing no properties beyond those which enter into its definition and their implications. We thus do not "learn" about a formal system, beyond establishing the consequences of our definitions through the application of conventional rules of inference, and sometimes by modifying or enlarging the initial definitions in particular ways. We have seen that even the study of formal systems is not free of strife and controversy; how much more strife can be expected when we attempt to relate this world to another of a fundamentally different character? And yet that is the task we now undertake; it is the basic task of relating *experiment* to *theory*. We shall proceed to develop a general framework in which formal and natural systems can be related, and then we shall discuss that framework in an informal way.

The essential step in establishing the relations we seek, and indeed the key to all that follows, lies in an exploitation of *synonymy*. We are going to force the name of a percept to be also the name of a formal entity; we are going to force the name of a linkage between percepts to also be the name of a relation between mathematical

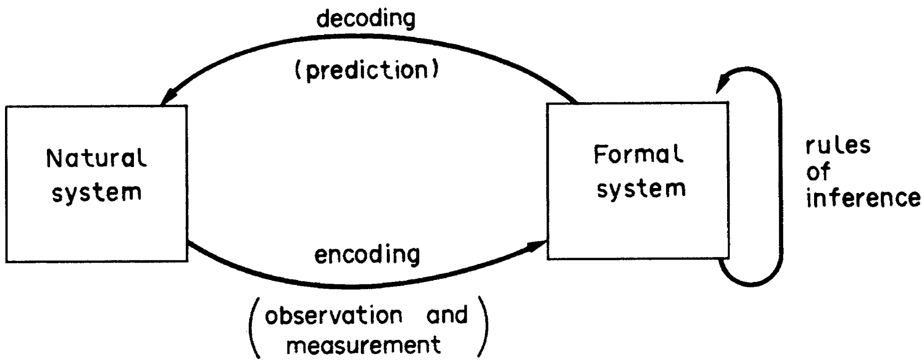


Fig. 2.1

entities; and most particularly, we are going to force the various temporal relations characteristic of causality in the natural world to be synonymous with the inferential structure which allows us to draw conclusions from premises in the mathematical world. We are going to try to do this in a way which is *consistent* between the two worlds; i.e. in such a way that the synonymies we establish do not lead us into contradictions between the properties of the formal system and those of the natural system we have forced the formal system to name. In short, we want our relations between formal and natural systems to be like the one Goethe postulated as between the genius and Nature: what the one promises, the other surely redeems.

Another way to characterize what we are trying to do here is the following: we seek to *encode* natural systems into formal ones in a way which is consistent, in the above sense. Via such an encoding, if we are successful, the inferences or theorems we can elicit within these formal systems become *predictions* about the natural systems we have encoded into them; consistency then means that these predictions will be *verified* in the natural world when appropriately *decoded* into linkage relations in that world. And as we shall see, once such a relation between natural and formal systems has been established, a host of other important relations will follow of themselves; relations which will allow us to speak precisely about analogy, similarity, metaphor, complexity, and a spectrum of similar concepts.

If we successfully accomplish the establishment of a relation of this kind between a particular natural system and some formal system, then we will obtain thereby a composite structure whose character is crudely indicated in Fig. 2.1 below:

In this figure, the arrows labelled "encoding" and "decoding" represent correspondences between the observables and linkages comprising the natural system and symbols or propositions belonging to the formal system. Linkages between these observables are also encoded into relations between the corresponding propositions in the formal system. As we noted earlier, the rules of inference of the formal system, by means of which we can establish new propositions of that system as implications, must be re-interpreted (or decoded) in the form of specific assertions pertaining to the observables and linkages of the natural system; these are the *predictions*. If the assertions *decoded* in this fashion are *verified* by observation; or what is the same thing, if the observed behavior of the natural system *encodes* into the same propositions as those obtained from the inferential rules of the formal system, we shall say that (to that extent) the relation between the two systems which we have established, and which is diagrammed in Fig. 2.1, is a *modeling relation*. Under these circumstances, we shall also say that the formal system of Fig. 2.1, modulo the encoding and decoding rules in question, is a model of the natural system to which it is related by those rules.

Let us consider the situation we have described in somewhat more detail. We saw in the preceding chapter that rules of inference relating propositions in a formal system cannot be applied arbitrarily, but only to propositions of a definite form or character. If we are going to apply our rules of inference in accordance with the diagram of Fig. 2.1, then specific aspects of the natural system must code propositions to which they can be applied. In the broadest terms, the aspects so encoded can be called *states* of the natural system. A state of a natural system, then, is an aspect of it which encodes into a *hypothesis* in the associated formal system; i.e. into a proposition of that formal system which can be the basis for an inference. Speaking very informally, then, *a state embodies that information about a natural system which must be encoded in order for some kind of prediction about the system to be made*. It should be explicitly noted that, according to this definition, *the concept of state of a natural system is not meaningful apart from a specific encoding of it into some formal system*. For, as we are employing the term, a state comprises what needs to be known about a natural system in order for other features of that system to be determined. This in itself is somewhat a departure from ordinary usage, in which a state of a natural system is regarded as possessing an objective existence in the external world. It should be compared with our employment of the term "state" in Sect. 1.2 above; we will see subsequently, of course, that the two usages in fact coincide.

Let us now suppose that we have encoded a state of our natural system into a hypothesis of the associated formal system. What kind of inferences can we expect to draw from that hypothesis, and how will they look when they are decoded into explicit predictions?

We shall begin our discussion of this question by recalling our discussion of *time* and its relation to percepts. We pointed out that there were two different aspects of our time-sense, which we called simultaneity and precedence, which we impose upon percepts. We then *impute* similar properties to the qualities in the external world which generate these percepts. Accordingly, it is reasonable to organize assertions about natural systems (and in particular, the predictions generated from a model) in terms of temporal features.

We can then recognize three temporally different kinds of predictions: (a) predictions which are time-independent; (b) predictions which relate different qualities of our system at the same instant of time; and (c) predictions which relate qualities of our system at different instants of time. Predictions of the first type are those which are unrelated to the manner in which time itself is encoded into our formal system. Typically, they express relations or linkages between qualities which play an essential role in the recognition of the identity of the natural system with which we are dealing, and hence which *cannot* change, without our perception that our original system has been replaced by a new or different system. To anticipate subsequent discussion somewhat, qualities of this type are exemplified by constitutive parameters; hence a time-independent prediction would typically take the form of a linkage between such constitutive parameters.

Almost by definition, all time-dependent qualities pertaining to our natural system can change without our perceiving that our system has been replaced by another. Hence, the two kinds of time-dependent predictions we might expect to make on the basis of a model will necessarily pertain to such qualities.

Predictions of the second type itemized above will generally be of the following form: if a state of our natural system at a given instant of time is specified, then some other quality pertaining to the system *at the same instant* may be inferred. That is, predictions of this type will *express linkages between system qualities which pertain to a specific instant of time*, and which are simultaneously expressed at that instant.

Predictions of the third type will generally be of the form: if the state of our natural system at a particular instant is specified, then some other quality pertaining to the system *at some other instant* may be inferred. Thus, predictions of this type *express linkages between qualities which pertain to different instants of time*. In particular, we may be interested in inferences which link *states* at a given instant to states at other instants. As we shall see, linkages of this type essentially constitute the dynamical laws of the system.

It will be noted that all three types of inferences, and the predictions about a natural system which may be decoded from them, express *linkage* relations between system qualities. The corresponding inferential rules in the formal system which decode into these linkage relations are what are usually called *system laws* of the natural system. It is crucial to note that, as was the case with the concept of state, the concept of system laws is meaningless apart from a specific encoding and decoding of our natural system into a formal one; the system laws belong to the formal system, and are in effect *imputed* to the associated natural system exactly by virtue of the encoding itself.

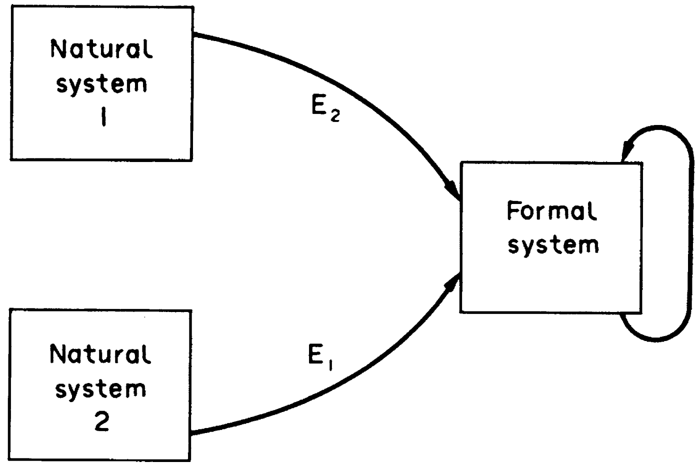
We shall see many specific examples of all of these ideas, in a broad spectrum of contexts, in Chap. 3 below; therefore we shall not pause to give examples here. However, the reader should at this point look back at the treatment of the reactive paradigm provided in Sect. 1.2 above, to gain an idea of just how much had to be tacitly assumed in writing down even the first line of that development.

We are now going to elaborate a number of important variations on the basic theme of the modeling relations embodied in Fig. 2.1 above. As we shall see, these variations represent the essential features of some of the most important (and hence perhaps, the most abused) concepts in theoretical science. They will all be of central importance to us in the developments of the subsequent chapters.

The first of these variations is represented in diagrammatic form in Fig. 2.2 below. In this situation, we have two natural systems, which we shall denote by N_1, N_2, encoded into the same formal system F, via particular encodings E_1, E_2, respectively.

The specific properties of the situation we have diagrammed depends intuitively on the degree of "overlap" in F between the set E_1 (N_1) of propositions which encode qualities of N_1, and the set $E_2(N_2)$ of propositions which encode qualities

Fig. 2.2

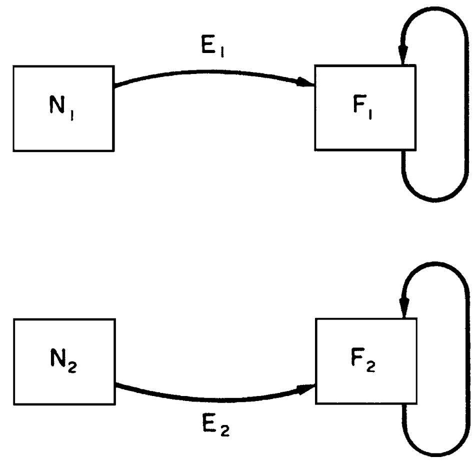


of N_2. Clearly, if the *same* proposition of F simultaneously encodes a quality of N_1 and a quality of N_2, *we may thereby establish a relation between* N_1 *and* N_2. It is convenient for us to recognize three cases:

Case A1: $E_1\,(N_1) = E_2(N_2)$

In this case, every proposition of F which encodes a quality of N_1 also encodes a quality of N_2, and conversely. Utilizing this fact, we may in effect construct a "dictionary", which allows us to *name* every quality of N_1 by that quality of N_2 which encodes into the same proposition in F; and conversely, every quality of N_2 can be named by a corresponding quality in N_1. Since by definition of the modeling relation itself, the system laws imputed to N_1 and N_2 are also the same, this dictionary also allows us to decode the inferences in F into corresponding propositions about N_1 and N_2 respectively. In this case, we can say that *the natural systems* N_1 *and* N_2 *share a common model.* Alternatively, we shall say that N_1 and N_2 are *analogous systems, or analogs.*[1] As usual, we point out explicitly that this concept of analogy is only meaningful with respect to a particular choice of the encodings E_1, E_2 into a common formal system F. A moment's reflection will reveal that our use of the term "analog" is a straightforward generalization of its employment in the term "analog computer". It must also be emphasized that the relation of analogy is a relation between *natural systems*; it is a relation established precisely when these natural systems share a common encoding into the same formal system. In this sense, it is different in character from the modeling relation, which (as we have defined it) is a relation between a natural system and a formal system. Despite this, we shall often abuse language, and say that N_1 and N_2 are *models of each other*; this is a usage which is ubiquitous in the literature, and can really cause no confusion. It should be noted that our "dictionary" relating corresponding qualities of analogous systems behaves like a *linkage* between them; but the origin of such a linkage has its roots in the formal system F and in the properties of the encodings E_1, E_2; it does *not* reside in the natural systems N_1, N_2 themselves.

Fig. 2.3



Case A2: $E_1(N_1) \subset E_2\,(N_2)$

This case differs from the previous one in that, while every proposition of F which encodes a quality of N_1 also encodes a corresponding property of N_2, the converse is not true. Thus if we establish a "dictionary" between the qualities of N_1 and N_2 put into correspondence by the respective encodings, we find that every quality of N_1 corresponds to some quality of N_2, but not conversely. That is, there will be qualities of N_2 which do not correspond in this fashion to any qualities in N_1.

Let N_2' denote those qualities of N_2 which can be put in correspondence with qualities of N_1 under these circumstances. If we restrict attention to just N_2', then we see that N_1 and N_2' are analogs, in the sense defined above. If we say that N_2' defines a *subsystem* of N_2, then the case we are discussing characterizes the situation in which N_1 and N_2 are not directly analogs, but in which N_2 *contains a subsystem analogous to* N_1. By abuse of language, we can say further that N_2 *contains a subsystem* N_2' *which is a model of* N_1.

Of course, if it should be the case that $E_2(N_2) \subset E_1(N_1)$, we may repeat the above argument, and say that N_1 possesses a subsystem N_1' which is analogous to, or a model of, N_2.

Case A3: $E_1\,(N_1) \cap E_2\,(N_2) \neq \emptyset$

We assume here that neither of the sets $E_1(N_1)$, $E_2(N_2)$ of encoded propositions is contained in the other, but that their intersection is non-empty. In this case, an application of the above argument leads us to define subsystems N_1' of N_1, and N_2' of N_2, which can be put into correspondence by virtue of their coding into the same proposition of F. In this case, then, we may say that the natural systems N_1, N_2 possess *subsystems* which are analogous, or which are models of each other.

There is of course a fourth possibility arising from the diagram in Fig. 2.3; namely $E_1(N_1) \cap E_2(N_2) = \emptyset$. But this possibility is uninteresting, in that the encodings E_1, E_2 allow us to establish no relation between N_1 and N_2.

Now let us consider a further variation on our basic theme, which generalizes the above discussion in a significant way. Let us consider the situation diagrammed in Fig. 2.3 below:

Here the natural systems N_1, N_2 encode into *different* formal systems F_1, F_2 respectively. We shall now attempt to establish relations between N_1 and N_2 on the basis of *mathematical* relations which may exist between the corresponding *formal* systems F_1, F_2, respectively.

Here again, it will be convenient to recognize a number of distinct cases, which are parallel to those just considered.

Case B1: $E_1(N_1)$ and E_2 (N_2) are *isomorphic*

In this case, there is a structure-preserving mapping between the sets of encoded propositions, which is one-to-one and onto. Since we are dealing here with formal systems, this mapping is a definite mathematical object, and can be discussed as such. By "structure- preserving", we mean here that this mapping can be extended in a unique way to all inferences in F_1, F_2 which can be obtained from $E_1(N_1)$, $E_2(N_2)$ respectively.

Under these circumstances, we can once again construct a dictionary between the qualities of N_1 and those of N_2, by putting qualities into correspondence whose encodings are images of one another under the isomorphism established between $E_1(N_1)$ and E_2 (N_2). This dictionary, and the fact that the system laws in both cases are also isomorphic, allows us to establish a relation between N_1 and N_2 which has all the properties of analogy defined previously. We shall call this relation an *extended analogy*, or, when no confusion is possible, simply an analogy, between the natural systems involved. It should be noted explicitly that the dictionary through which an extended analogy is established depends on the *choice* of isomorphism between E_1 (N_1) and $E_2(N_2)$.

Case B2: E_1 (N_1) is isomorphic to a subset of $E_2(N_2)$

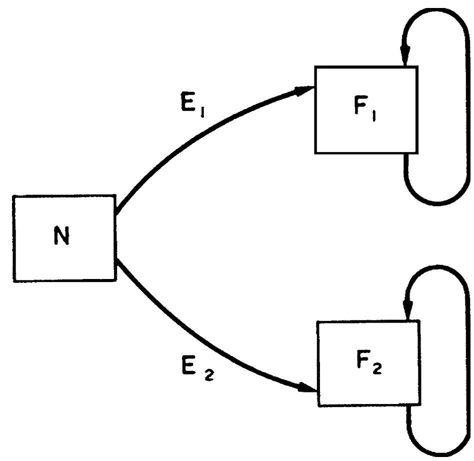
This case is, of course, analogous to Case A2 above. As in that case, we can establish our dictionary between N_1 and a subsystem N_2' of N_2, and thereby establish a relation of extended analogy between N_1 and N_2'. It is clear that all aspects of our discussion of Case A2 also apply here. Likewise, we have

Case B3: A subset of E_1 (N_1) is isomorphic to a subset of E_2 (N_2)

This case is the analog of Case A3 above, we can establish an extended analogy between subsystems N_1' of N_1 and N_2' of N_2 in the obvious fashion.

There is a final case to be considered before leaving the situation diagrammed in Fig. 2.3, which should be mentioned at this time, but about which we will not be able to be very precise until somewhat later. This is the circumstance in which the formal systems F_1, F_2 of that diagram satisfy some more general mathematical relation than that embodied in the existence of a structure-preserving mapping between them. As we shall see subsequently, such a relation will typically involve the satisfaction of some general criterion according to which formal systems may be classified; as for instance that F_1, F_2 are both "finite-dimensional" or "finitely generated" in some sense, but without any more detailed relationship which might be embodied in the existence of a structure-preserving mapping between them. In such a circumstance, we cannot establish a detailed dictionary relating qualities of the associated natural

Fig. 2.4



systems N_1, N_2. However, we can utilize the fact that F_1 and F_2 are related in some more general sense, and hence are to be counted as alike in that sense, to *impute* a corresponding relation to N_1, N_2 themselves. The situation we are describing will provide the basis for saying that N_1 is a *metaphor* for N_2, in the sense that *their models share some common general property*. Subsequently we shall see many examples of such metaphors, as for instance in the assertion that open systems are metaphors for biological developmental processes. This concept of metaphor will be important to us, but we will leave the more detailed discussion for subsequent chapters.

So far, we have considered situations in which different natural systems can be encoded into the same formal system, or into systems between which some mathematical relation can be established. We will now turn to the dual situation, in which the same natural system can be encoded into two (or more) formal systems; this situation is diagrammed in Fig. 2.4 below:

We will now take up the important question of the extent to which a mathematical relation can be established between the *formal* systems F_1 and F_2, simply by virtue of the fact that the same natural system N can be encoded into both of them. Such a relation would essentially embody a *correspondence principle* between the encodings.

In a sense, the diagram of Fig. 2.4 can be regarded as a variant of that shown in Fig. 2.3, in which we put both N_1 and N_2 equal to N. Indeed, if we consider the simplest situation, in which exactly the same variates of N are encoded into F_1 and F_2 (via the encodings E_1 and E_2 respectively), then it is easy to see that we are indeed in the situation previously discussed, and in fact essentially in the Case B1. However, we shall ignore this relatively trivial situation, and instead consider what happens when *different qualities* of N are encoded into F_1 and F_2. We are now in a quite different situation from that diagrammed in Fig. 2.3, for the following reason: we supposed that N_1 and N_2 represent *different* natural systems; this means that their respective qualities are generally unlinked. In Fig. 2.4, on the other hand, even though we may generally encode different qualities of N into different formal systems F_1, F_2, these qualities are linked in N through the hypothesis that they pertain to the *same* system. However, the linkages between these differently encoded qualities are themselves encoded neither in F_1 nor in F_2. Thus the question we are now posing is the following: *to what extent can these uncoded linkages manifest themselves in the form of purely mathematical relations between* F_1 *and* F_2? It will be seen that this is the inverse of the former cases, in which we *posited* a mathematical relation between F_1 and F_2, and *imputed* it to the natural systems encoded into them. Stated otherwise: whereas we formerly sought a relation between natural systems through a mathematical relation between their encodings, we presently seek a mathematical relation between encodings, arising from uncoded linkages between families of differently encoded qualities.

There are again a number of cases which must be considered separately.

Case C1

Let us suppose that we can establish a mapping of F_1, say, into F_2. That is, every proposition of F_1 can be associated with a unique proposition of F_2, but we do not suppose that the reverse is the case. Then every quality of N encoded into F_1 can thereby be associated with another quality of N, encoded into F_2, but not conversely. There is thus an important sense in which the encoding F_1 is *redundant*; every quality encoded into F_1 can be recaptured from the encoding of apparently different qualities into F_2. In this case, we would say intuitively that F_2 provides the more comprehensive formal description of N, and in fact that the encoding into F_1 can be *reduced* to the encoding into F_2.

The fundamental importance of this situation lies in its relation to *reductionism* as a basic scientific principle.[2] Reductionism asserts essentially that there is a universal way of encoding an arbitrary natural system, such that *every other encoding* is reducible to it in the above sense. It must be stressed, and it is clear from the above discussion, that reductionism involves a *mathematical* relation between certain formal systems, into which natural systems are encoded. In order for reductionism in this sense to be established as a *valid* scientific principle, the following data must be given: (a) a recipe, or algorithm, for producing the universal encoding for any given natural system N; (b) a *mathematical* proof that every other coding can be effectively mapped into the universal one. Needless to say, no arguments bearing on these matters have ever been forthcoming, and as we shall see abundantly, there are good reasons to suppose that the reductionistic principle is in fact false. Furthermore, in a purely practical sense, an attempt to obtain specific inferences about qualities of a given natural system N by detouring through a universal description would not be of much value in any case. Nevertheless, it is important to determine in what circumstances a given encoding E_1 (N) can be reduced to another one E_2 (N). For such a reduction implies important linkages between the qualities encoded thereby; linkages which as we noted above are themselves not explicitly encoded by either E_1 or E_2.

Case C2

Here we shall suppose that a mapping can be established from a *subset* of F_1 to a *subset* of F_2, in such a way that certain qualities of N encoded into $E_1(N_1)$ can be related to corresponding qualities encoded into E_2 (N_2). The mapping itself, of course, establishes a logical relation between the encodings, but one which does not hold universally. Thus in this case we may say that a *partial* reduction is possible, but one of limited applicability. As we shall see, when we leave the domain in which such a partial reduction is valid, the two encodings become logically independent; or what is the same thing, the linkage between the qualities in N which are encoded thereby is broken. Depending on the viewpoint which is taken, such a departure from the domain in which reduction is valid will appear as a *bifurcation*, or as an *emergence phenomenon*; i.e. the generation of an emergent novelty. We will develop explicit examples of these situations subsequently.

Case C3

In this case, no mathematical relation can be established between F_1 and F_2. Thus, these encodings are inadequate in principle to represent any linkages in N between the families or qualities encoded into them. Into this class, we would argue, fall the various *complementarities* postulated by Bohr in connection with microphysical phenomena. Here again, it must be stressed that it is the mathematical character of the encodings which determines whether or not relations between them can be effectively established; in this sense, complementarity is entirely a property of formal systems, and not of natural ones.

One further aspect of the basic diagram of Fig. 2.4 may be mentioned briefly here. We are subsequently going to relate our capacity to produce independent encodings of a given natural system N with the *complexity*[3] of N. Roughly speaking, the more such encodings we can produce, the more complex we will regard the system N. Thus, contrary to traditional views regarding system complexity, we do not treat complexity as a property of some particular encoding, and hence identifiable with a mathematical property of a formal system (such as dimensionality, number of generators, or the like). Nor is complexity entirely an objective property of N, in the sense of being itself a directly perceptible quality, which can be measured by a meter. Rather, complexity pertains at least as much to *us as observers* as it does to N; it reflects *our ability to interact with* N in such a way as to make its qualities visible to us. Intuitively speaking, if N is such that we can interact with it in only a few ways, there will be correspondingly few distinct encodings we can make of the qualities which we perceive thereby, and N will appear to us as a *simple* system; if N is such that we can interact with it in many ways, we will be able to produce correspondingly many distinct encodings, and we will correspondingly regard N as complex. It must be recognized that we are speaking here entirely of complexity as an attribute of *natural* systems; the same word (complexity) may, and often is, used to describe some attribute of a *formal* system. But this involves quite a different concept, with which we are not presently concerned. We shall also return to these ideas in more detail in subsequent chapters.

Let us turn now to an important re-interpretation of the various situations we have discussed above. We have so far taken the viewpoint that the modeling relation exemplified in Fig. 2.1 involves the encoding of a given natural system N into an appropriate formal system F, through an identification of the names of qualities of N with symbols and propositions in F. That is, we have supposed that we *start* from a given natural system whose qualities have been directly perceived in some way, and that the encoding is established *from* N *to* F. There are many important situations, however, in which the same diagram is established from precisely the opposite point of view. Namely, suppose we want to *build*, or create, a natural system N, in such a way that N possesses qualities linked to one another in the same way as certain propositions in some formal system F. In other words, we want the system laws characterizing such a system N to represent, or *realize*, rules of inference in F. This is, of course, the problem faced by an engineer or an architect who must *design* a physical system to meet given specifications. He invariably proceeds by creating first a *formal system*, a blueprint, or plan, according to which his system N will actually be built. The relation between the system N so constructed, and the plan from which it is constructed, is precisely the modeling relation of Fig. 2.1, but now the relation is generated *in the opposite direction* from that considered above.[4] From this we see that the analytic activities of a scientist seeking to understand a particular unknown system, and the synthetic activities of an engineer seeking to fabricate a system possessing particular qualities, are in an important sense inverses of one another. When we speak of the activities of the scientist, we shall regard the modeling relations so established in terms of the *encoding* of a natural system into a formal one; when we speak of the synthetic activities of the engineer, we shall say that the natural system N in question is a *realization* of the formal system F, which constitutes the plan. Using this latter terminology, then, we can re-interpret the various diagrams presented above; for instance, we can say that two natural systems N_1, N_2 are analogs precisely to the extent that they are *alternate realizations* of the same formal system F. Likewise, in the situation diagrammed in Fig. 2.4, we can say that the same natural system N can simultaneously realize two distinct formal systems; the complexity of N thus can be regarded as the number of distinct formal systems which we perceive that N can simultaneously realize. We shall consider these ideas also in more detail in the course of our subsequent exposition.

To conclude these introductory heuristic remarks regarding the modeling relation, let us return again to the situations diagrammed in Figs. 2.2 and 2.3 above. We saw in our discussion of those diagrams that the modeling relation is one established between a natural system N and a formal system F through a suitable encoding of qualities of N into symbols and propositions in F. We also saw that when two distinct natural systems N_1, N_2 can be appropriately encoded into the same formal system, it was meaningful to speak of a relation of *analogy* as being established between the natural systems N_1 and N_2 themselves. We wish to pursue a few corollaries of these ideas.

The main property of a modeling relation, which we shall exploit relentlessly in what follows, is this: once such a relation has been established (or posited) between a natural system N and a formal system F, *we can learn about* N *by* *studying* F; a patently different system. Likewise, once an analogy relation has been established between two natural systems N_1 and N_2, *we can learn about one of them by studying the other*. In this form, both modeling and analogy have played perhaps the fundamental role in both theoretical and experimental science, as well as a dominant role in every human being's daily life. However, many experimental scientists find it counter-intuitive, and therefore somehow outrageous, to claim to be learning about a particular natural system N_1 by studying a different system N_2, or by studying some formal system F, which is of a fundamentally different character from N_1. To them, it is self-evident that the only way to learn about N_1 is to study N_1; any other approach is at best misguided and at worst fraudulent. It is perhaps worthwhile, therefore, to discuss this attitude somewhat fully.

It is not without irony that the empiricist's aversion to models finds its nourishment in the same soil from which our concept of the modeling relation itself sprang. For they are both products of the development which culminated in Newton's *Principia*. In that epic work, whose grandeur and sweep created reverberations which still permeate so much of today's thought, it was argued that a few simple formal laws could account for the behavior of all systems of material particles, from galaxies to atoms. We have sketched a little of this development in Sect. 1.2 above. Newton also provided the technical tools required for these purposes, with his creation of the differential and integral calculus; i.e. he also provided the formal basis for encoding the qualities of particulate systems, and drawing conclusions or inferences pertaining to those systems from the system laws. Not only was the entire Newtonian edifice one grand model in itself, but it provided a recipe for establishing a model of *any* particulate system. To set this machinery in motion for any specific system, all that was needed was (a) a specification of the forces acting on the system (i.e. an encoding of the system laws), and (b) a knowledge of initial conditions.

The universality of the Newtonian scheme rests on the extent to which we can regard *any* natural system as being composed of material particles. If we believe that any system can in fact be resolved into such particles, then the Newtonian paradigm can always be invoked, and the corresponding machinery of encoding and prediction set in motion. The belief that this is always possible (and more than this, is always necessary and sufficient) is the motivating force behind the concept of reductionism, which was mentioned earlier.

Now the determination of system laws, and even more, the specification of initial conditions, which are the prerequisites for the application of the Newtonian paradigm, are not themselves conceptual or theoretical questions; they are empirical problems, to be settled by observation and measurement, carried out on the specific system of interest. Thus, hand in hand with the grandest of conceptual schemes we necessarily find the concomitant growth of empiricism and reductionism; two thankless offspring which incessantly threaten to devour that which gave them birth. For, once separated from the conceptual framework which originally endowed them with meaning and significance, they become in fact antithetical to that framework. Pursued in isolation, empiricism must come to claim (a) that measurement and observation are ends in themselves; (b) that measurement and observation pertain only to extremely limited and narrow families of interactions which we can generate in the here-and-now; (c) and that, finally, whatever is not the result of measurement and observation in this narrow sense *does not exist*, and ergo cannot be part of science.

We can now begin to understand why the theoretical concept of a model, which has as its essential ingredient a freely posited formal system, is so profoundly alienating to those reared in an exclusively empirical tradition. This alienation can sometimes take bizarre and even sinister forms. Let us ponder, for example, the growth of what was called *Deutsche Physik*, which accompanied the rise of National Socialism in Germany not so very long ago.[5] At root, this was simply a rebellion on the part of some experimental physicists against a threatening new theoretical framework for physics (especially as embodied in general relativity); the ugly racial overtones of this movement were simply an indication of how desperate its leaders had become in the face of such a challenge. They were reduced to arguing that "those who studied equations instead of nature" were traitors to their abstract ideal of science; that the true scientist could only be an experimenter who developed a kind of mystic rapport with nature through direct interaction with it. Similarly, the phenomenon now called Lysenkoism in the Soviet Union was associated with a phobia against theoretical developments in biology, especially in genetics.[6] These are admittedly extreme situations, but they are in fact the logical consequences of equating science with empiricism (which in itself, it should be noted, involves a model).

Although perhaps most empirical scientists would dissociate themselves from such aberrations, they tend to share the disquiet which gave rise to them. They regard the basic features of the modeling relation, and the idea that we can learn about one system by studying another, as a reversion to magic; to the idea that one can divine properties of the world through symbols and talismans. Perhaps it must inevitably seem so to those who think that science is done when qualities are simply named (as meter readings) and tabulated. But in fact, that is the point at which science only begins.

From this kind of acrimony and strife, one might suppose that the modeling relation is something arcane and rare. We have already seen that this is not so; that indeed the modeling relation is a ubiquitous characteristic of everyday life as well as of science. But we have not yet indicated just how widespread it is. To do so, and at the same time to give specific instances of the situations sketched abstractly in our discussion so far, we will now turn to developing such instances, as they appear in many fields. We may in fact regard such instances as *existence proofs*, showing that the various situations we have diagrammed above are not vacuous. This will be the primary task of the chapters in Chap. 3 to follow.

References and Notes

1. Aside from the special case of "dynamical similarity", which we shall consider at some length in Sect. 3.3 (especially Example 5), there has still been no systematic

study of the circumstances under which two different systems are analogous. Our treatment here is based on ideas first expounded in a paper of the author's: Rosen, R., *Bull. Math. Biophys*. 30, 481–492 (1968).
2. See also Sect. 3.6 below.
3. See Sect. 5.7 below.
4. The relation between model and realization is discussed more fully in Sect. 3.1.
5. For a good discussion of the philosophical issues involved here, see Beyerchen, A. D., *Scientists Under Hitler*. Yale University Press (1977).
6. The two definitive treatments of Lysenkoism are Medvedev, Zh. A., *The Rise and Fall of T. D. Lysenko*. Columbia University Press (1969).

   Joravsky, D., *The Lysenko Affair*. Harvard University Press (1970).
   These three books make fascinating reading, especially when read concurrently.